VF Asia Footwear Laboratory Testing Manual

Test Method for Water Wicking	Test method # VF- E14 REF # : TM-R- 056 / TM-R-021 Page 1 of 5
Date Issued : July 11, 2014 Revision # : 3 Reason for Revisions : Added TBL requirement for Non-Gore Tex construction	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To test the water wicking behavior of footwear components, including textiles, leather, synthetic leather, laces, webbings, threads, zipper, foams, heel counters and other footwear components.

REFERENCE: DIN 53924

EQUIPMENT AND MATERIAL:

1. Wicking tray with Gripper Set and Flume
2. Stopwatch or timer with an accuracy of +/-0.5 seconds.
3. Dye preferably Solophenyl Turquoise BRLE 400%
4. Paper clip or other type of sinker weights, which can be clamped onto one end of the test material ,so that the test sample will remain immersed in the water
5. White paper or paper towel
6. Steel scale, capable of measuring to 1.0mm.
7. Pre-treatment equipment for ***Gore Tex construction :***
   - Maeser Flex
   - Leather Sewing Machine
   - Lace Abrasion Tester
   - Load Weight of 1,500 g (in the absence of Lace Abrasion Tester)
8. Fabrics, laces, webbings, threads, zipper, foams, heel counters

MATERIAL PREPARATION:

Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

Option 1: No pre –stressing (For Non- Gore Tex construction).

Option 2: Pre-STRESSING of COMPONENTS (For Gore Tex construction)

PRE-STRESSING PROCEDURES:

A. Fabric , Synthetic Leathers , Bound Textile and Bound Leather

1. Cut sample size 114+/-2 mm by 114 +/-2 mm for Maeser flexing in each direction (warp and weft) and label.
2. Clamp top and bottom of sample into machine. Face side towards water.
3. Submerge sample 2.5 cm from the bottom of the folded specimen and also place water inside of sample at 2.5 cm deep.

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4. Run for 3,000 flexes, approximately 30 minutes.
5. After Maeser flexing is complete, let sample air dry for at least 12 hrs before cutting the specimens for wicking test.
6. Take samples from the center of each flexed piece. Cut to the required dimension .

Thread:

1. Pre- stitch the thread on a piece of 1.6mm – 2.2 mm Full Grain Leather (free from fat and don't use any oil). Use 3 stitches /cm -high speed, without intermediate stop). Use R-needles
2.

Thread Size	Needle Size
11/3	140
20/3	130
30/3	120
40/3	110
60/3	100
75/3	90

3. After stitching rip the lower suture line using seam ripper / scalpel and remove upper thread.
4. Sample for wicking is taken from this upper thread.
5. Attach sinker onto end of each thread for testing.

B. Lace / Webbing :

Using the Lace Abrasion Tester:

1. Open the lid of the machine and make sure that for laces the machine is equipped with turn -round for laces (round eye) and for webbing the machine is equipped with rectangular shaped slot.

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2. Push the start button to position slider to the side needed to test lace or webbing. Slider should be positioned to the side where the weight is placed
3. First fix the lace/webbing at the weight and put the weight back at their position (into the tube), then pass the lace / webbing through the first eyelet and then through the second eyelet.
4. Pass the end of the lace /webbing through the fixing device.
5. Push at the lace/webbing until the weight gets lifted up (no contact to the bottom and fix the lace /webbing with screw).
6. Using a pen make a mark behind the second eyelet.
7. Put each individual counter, for each specimen to “0” and turn on each individual counter at each used testing position.
8. Attached sinker onto end of each lace for testing if necessary.

Using a “Load Weight “of 1,500g.

1. Thread the remaining lace/ webbing through the load weight and allow a back and forward motion (200 revolutions)
2. Lace should be held at 45 ° angle.
3. Samples are taken from the pre-stressed lace by measuring 7.5cm. Ensure that the stressed area is used for the wicking test.

SAMPLE PREPARATION:

A. Fabric , Synthetic Leathers , Bound Textile and Bound Leather (Note : No pre-stressing for regular leather , non – bound foams or zippers)

1. Take samples from the center of each flexed piece – pre-stressed from Maeser
2. Die or hand cut a sample which is 25+/-1mm by 75+/-5mm.The direction of the fabric should be noted (the length direction has the end yarns running lengthwise).
3. Draw two lines across the test sample, one line is located 10+/-1mm from one end of the test sample, and the second line is located 10 +/-1mm above the first line.

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B. Thread

1. Take samples which are taken from upper thread – pre-stressed from sewn upper in leather. Length dimension is 75+/-5mm long.
2. Attach sinker / weight onto end of each thread for testing so that the test sample will remain immersed in the water.
3. Draw two lines across the test samples. One line is located 10+/-1mm from one end of the test sample, and the second line is located 10 +/-1mm above the first line.

C. **Lace / Webbing :**

1. Take sample taken from the pre-stressed lace by measuring 7.5cm. Ensure that the stressed area is used for the wicking test.

2. Attach sinker/ weight onto end of each lace if necessary so that the test sample will remain immersed in the water.
3. Draw two lines across the test samples. One line is located 10+/-1mm from one end of the test sample, and the second line is located 10 +/-1mm above the first line.

D. **Leather :**

1. Cut samples 2.5 cm x 7.5 cm, warp and weft directions (Note: Leather sample for wicking should not be pre-stressed.)

2. Draw two lines across the test samples. One line is located 10+/-1mm from one end of the test sample, and the second line is located 10 +/-1mm above the first line *(Note : For TBL WP Construction : 13 +/-1mm above the first line) .*

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Fill the wicking tray with distilled water at 23+/-2ºC to a depth of approximately 22 ±3 mm.

Note: Depending on the color of the test sample, it may be useful to add dye to the water so that it can be more easily seen as it wicks up the test sample

2. For thread and lace attach the sinker / weights to the test samples so that it will remain immersed in the water.
3. Fix the specimen with the gripper set. Then put the gripper set on a tester.
4. Fix the specimen into the flume and check waterline. Ensure the first 10 mm line on sample is at the water level.

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5. Immediately upon immersing the test sample, start the stopwatch for ***1 hour test for option 1 – TBL requirement / Non- Gore Tex construction and 2 hours for option 2 / Gore Tex Construction.***.
6. After **1 hour or 2 hours,** remove the material and carefully put on white paper or paper towel
7. Press the specimen (front side) from top to bottom by finger for water marking on white paper or paper towel. Pressing should not exceed the first 10 mm line.
8. Press the side for water marking on white paper or paper towel. Press the specimen (back side) top to bottom – refer step # 7.
9. Check and measure the watermark.

REPETITIONS: Test at least two (2) samples in each direction per material

REPORT: Height of water rises up as specified